

2.21

Question: Registrants at a large convention are offered 6 sightseeing tours on each of 3 days. In how many ways can a person arrange to go on a sightseeing tour planned by this convention?

Solution:

A person has 6 choices of tours for each of the 3 days. Therefore, the total number of ways is:

$$6 \times 6 \times 6 = 6^3 = 216$$

Answer: 216

2.22

Question: In a medical study, patients are classified in 8 ways according to whether they have blood type AB^+ , AB^- , A^+ , A^- , B^+ , B^- , O^+ , or O^- , and also according to whether their blood pressure is low, normal, or high. Find the number of ways in which a patient can be classified.

Solution:

There are 8 blood types and 3 blood pressure categories. Using the multiplication rule:

$$8 \times 3 = 24$$

Answer: 24

2.23

Question: If an experiment consists of throwing a die and then drawing a letter at random from the English alphabet, how many points are there in the sample space?

Solution:

A die has 6 faces, and the English alphabet has 26 letters. The total number of outcomes is:

$$6 \times 26 = 156$$

Answer: 156

2.24

Question: Students at a private liberal arts college are classified as being freshmen, sophomores, juniors, or seniors, and also according to whether they are male or female. Find the total number of possible classifications for the students of that college.

Solution:

There are 4 class levels (freshman, sophomore, junior, senior) and 2 genders (male, female). The total number of classifications is:

$$4 \times 2 = 8$$

Answer: 8



2.25

Question: A certain brand of shoes comes in 5 different styles, with each style available in 4 distinct colors. If the store wishes to display pairs of these shoes showing all of its various styles and colors, how many different pairs will the store have on display?

Solution:

For each of the 5 styles, there are 4 colors. Thus, the total number of pairs is:

$$5 \times 4 = 20$$

Answer: 20

2.26

Question: A California study concluded that following 7 simple health rules can extend a man's life by 11 years on average and a woman's life by 7 years. These 7 rules are as follows: no smoking, get regular exercise, use alcohol only in moderation, get 7 to 8 hours of sleep, maintain proper weight, eat breakfast, and do not eat between meals. In how many ways can a person adopt 5 of these rules to follow:

- (a) if the person presently violates all 7 rules?
- (b) if the person never drinks and always eats breakfast?

Solution:

(a) If the person violates all 7 rules, choosing 5 rules to follow is equivalent to choosing 5 items from 7. This is a combination problem:

$$\binom{7}{5} = \frac{7!}{5!(7-5)!} = \frac{7 \times 6}{2 \times 1} = 21$$

(b) If the person never drinks and always eats breakfast, these two rules are fixed. The person must choose 3 additional rules from the remaining 5:

$$\binom{5}{3} = \frac{5!}{3!(5-3)!} = \frac{5 \times 4}{2 \times 1} = 10$$

Answers:

(a) 21

(b) 10

2.27

Question: A developer of a new subdivision offers a prospective home buyer a choice of 4 designs, 3 different heating systems, a garage or carport, and a patio or screened porch. How many different plans are available to this buyer?

Solution:

The total number of plans is the product of the choices:

$$4 \times 3 \times 2 \times 2 = 48$$

Answer:

2.28

Question: A drug for the relief of asthma can be purchased from 5 different manufacturers in liquid, tablet, or capsule form, all of which come in regular and extra strength. How many different ways can a doctor prescribe the drug for a patient suffering from asthma?

Solution:

There are 5 manufacturers, 3 forms (liquid, tablet, capsule), and 2 strengths (regular, extra). The total number of prescriptions is:

$$5 \times 3 \times 2 = 30$$

Answer:

2.29

Question: In a fuel economy study, each of 3 race cars is tested using 5 different brands of gasoline at 7 test sites located in different regions of the country. If 2 drivers are used in the study, and test runs are made once under each distinct set of conditions, how many test runs are needed?

Solution:

The total number of test runs is:

$$3 \times 5 \times 7 \times 2 = 210$$

Answer:

2.30

Question: In how many different ways can a true-false test consisting of 9 questions be answered?

Solution:

Each question has 2 possible answers (true or false). For 9 questions:

$$2^9 = 512$$

Answer:

2.31

Question: A witness to a hit-and-run accident told the police that the license number contained the letters RLH followed by 3 digits, the first of which was a 5. If the witness cannot recall the last 2 digits, but is certain that all 3 digits are different, find the maximum number of automobile registrations that the police may have to check.

Solution:

The first digit is fixed as 5. The second digit can be any of the remaining 9 digits (0–9 except 5), and the third digit can be any of the remaining 8 digits. Thus:

$$9 \times 8 = 72$$

Answer:

2.32

Question:

- (a) In how many ways can 6 people be lined up to get on a bus?
- (b) If 3 specific persons, among 6, insist on following each other, how many ways are possible?
- (c) If 2 specific persons, among 6, refuse to follow each other, how many ways are possible?

Solution:

- (a) The total number of arrangements of 6 people is:

$$6! = 720$$

- (b) Treat the 3 specific people as a single "block." There are $4!$ ways to arrange the block and the other 3 people, and $3!$ ways to arrange the people within the block:

$$4! \times 3! = 24 \times 6 = 144$$

- (c) First, calculate the total arrangements where the 2 specific people are together. Treat them as a single "block," giving $5!$ arrangements of the block and the other 4 people, and $2!$ arrangements within the block:

$$5! \times 2! = 120 \times 2 = 240$$

Subtract this from the total arrangements:

$$6! - 240 = 720 - 240 = 480$$

Answers:

(a)

(b)

(c)



2.33

Question:

- (a) In how many different ways can a student check off one answer to each question on a multiple-choice test consisting of 5 questions, each with 4 possible answers?
- (b) In how many ways can a student check off one answer to each question and get all the answers wrong?

Solution:

- (a) Each question has 4 choices. For 5 questions:

$$4^5 = 1024$$

- (b) To get all answers wrong, the student must avoid the correct answer for each question. For each question, there are 3 incorrect choices. Thus:

$$3^5 = 243$$

Answers:

- (a)
- (b)

2.34

Question:

- (a) How many distinct permutations can be made from the letters of the word COLUMNS?
- (b) How many of these permutations start with the letter M?

Solution:

- (a) The word "COLUMNS" has 7 unique letters. The total number of permutations is:

$$7! = 5040$$

- (b) Fix the first letter as M. The remaining 6 letters can be arranged in:

$$6! = 720$$

Answers:

- (a)
- (b)

2.35

Question: A contractor wishes to build 9 houses, each different in design. In how many ways can he place these houses on a street if 6 lots are on one side of the street and 3 lots are on the opposite side?

Solution:

First, choose 6 houses for one side of the street ($\binom{9}{6}$) and arrange them. Then arrange the remaining 3 houses on the other side:

$$\binom{9}{6} \times 6! \times 3! = 84 \times 720 \times 6 = 362,880$$

Answer:

2.36

Question:

- (a) How many three-digit numbers can be formed from the digits 0, 1, 2, 3, 4, 5, and 6 if each digit can be used only once?
(b) How many of these are odd numbers?
(c) How many are greater than 330?

Solution:

(a) The first digit cannot be 0. Choose the first digit from {1, 2, 3, 4, 5, 6} (6 choices). The second digit can be any of the remaining 6 digits, and the third digit can be any of the remaining 5 digits:

$$6 \times 6 \times 5 = 180$$

(b) For the number to be odd, the last digit must be {1, 3, 5}. Choose the last digit (3 choices), the first digit (5 choices, excluding 0 and the last digit), and the middle digit (5 choices):

$$3 \times 5 \times 5 = 75$$

(c) Numbers greater than 330 must either start with 3 and have a second digit > 3 , or start with > 3 .

Case 1: Start with 3. Second digit > 3 : {4, 5, 6} (3 choices). Third digit: 5 choices:

$$3 \times 5 = 15$$

Case 2: Start with > 3 : {4, 5, 6} (3 choices). Remaining digits: $6 \times 5 = 30$:

$$3 \times 30 = 90$$

Total:

$$15 + 90 = 105$$

Answers:

- (a)
(b)
(c)



2.37

Question: In how many ways can 4 boys and 5 girls sit in a row if the boys and girls must alternate?

Solution:

The arrangement must alternate starting with either a boy or a girl.

Case 1: Start with a boy. Arrange the boys (4!) and the girls (5!):

$$4! \times 5! = 24 \times 120 = 2880$$

Case 2: Start with a girl. Similarly:

$$5! \times 4! = 120 \times 24 = 2880$$

Total:

$$2880 + 2880 = 5760$$

Answer:

2.38

Question: Four married couples have bought 8 seats in the same row for a concert. In how many different ways can they be seated:

- (a) with no restrictions?
- (b) if each couple is to sit together?
- (c) if all the men sit together to the right of all the women?

Solution:

- (a) No restrictions:

$$8! = 40,320$$

- (b) Treat each couple as a "block." Arrange the 4 blocks ($4!$) and arrange the individuals within each block ($2!$ per block):

$$4! \times (2!)^4 = 24 \times 16 = 384$$

- (c) Arrange the women ($4!$) and the men ($4!$):

$$4! \times 4! = 24 \times 24 = 576$$

Answers:

- (a)

40,320

- (b)

384

- (c)

576

2.39

Question: In a regional spelling bee, the 8 finalists consist of 3 boys and 5 girls. Find the number of sample points in the sample space S for the number of possible orders at the conclusion of the contest for:

- (a) all 8 finalists;
- (b) the first 3 positions.

Solution:

- (a) Total arrangements of 8 finalists:

$$8! = 40,320$$

- (b) Choose the first 3 positions from 8 finalists:

$$P(8, 3) = \frac{8!}{(8-3)!} = 8 \times 7 \times 6 = 336$$

Answers:

- (a)

40,320

 - (b)

336

-

2.40

Question: In how many ways can 5 starting positions on a basketball team be filled with 8 men who can play any of the positions?

Solution:

Choose 5 players from 8 and arrange them:

$$P(8, 5) = \frac{8!}{(8-5)!} = 8 \times 7 \times 6 \times 5 \times 4 = 6720$$

Answer: 6720

2.41

Question: Find the number of ways that 6 teachers can be assigned to 4 sections of an introductory psychology course if no teacher is assigned to more than one section.

Solution:

Assign 4 teachers to the sections ($P(6, 4)$):

$$P(6, 4) = \frac{6!}{(6-4)!} = 6 \times 5 \times 4 \times 3 = 360$$

Answer: 360

2.42

Question: Three lottery tickets for first, second, and third prizes are drawn from a group of 40 tickets. Find the number of sample points in S for awarding the 3 prizes if each contestant holds only 1 ticket.

Solution:

Choose and arrange 3 winners from 40:

$$P(40, 3) = \frac{40!}{(40-3)!} = 40 \times 39 \times 38 = 59,280$$

Answer: 59,280

2.43

Question: In how many ways can 5 different trees be planted in a circle?

Solution:

In a circular arrangement, fix one tree and arrange the remaining 4:

$$(5-1)! = 4! = 24$$

Answer: 24

2.44

Question: In how many ways can a caravan of 8 covered wagons from Arizona be arranged in a circle?

Solution:

Fix one wagon and arrange the remaining 7:

$$(8 - 1)! = 7! = 5040$$

Answer: 5040

2.45

Question: How many distinct permutations can be made from the letters of the word INFINITY?

Solution:

The word "INFINITY" has 8 letters, including 3 I's, 2 N's, and 1 each of F, T, Y. The total number of permutations is:

$$\frac{8!}{3! \times 2!} = \frac{40,320}{6 \times 2} = 3360$$

Answer: 3360

2.46

Question: In how many ways can 3 oaks, 4 pines, and 2 maples be arranged along a property line if one does not distinguish among trees of the same kind?

Solution:

The total number of trees is $3 + 4 + 2 = 9$. The number of arrangements is:

$$\frac{9!}{3! \times 4! \times 2!} = \frac{362,880}{6 \times 24 \times 2} = 1260$$

Answer: 1260

2.47

Question: How many ways are there to select 3 candidates from 8 equally qualified recent graduates for openings in an accounting firm?

Solution:

This is a combination problem:

$$\binom{8}{3} = \frac{8!}{3!(8-3)!} = \frac{8 \times 7 \times 6}{3 \times 2 \times 1} = 56$$

Answer: 56

2.48

Question: How many ways are there that no two students will have the same birth date in a class of size 60?

Solution:

There are 365 days in a year. The first student has 365 choices, the second has 364, and so on. The total number of ways is:

$$P(365, 60) = \frac{365!}{(365 - 60)!}$$

This value is extremely large and typically computed using software.

Answer: $P(365, 60)$